National
Qualifications
2024

X819/76/11 Design and Manufacture

WEDNESDAY, 1 MAY
1:00 PM – 3:15 PM

Total marks — 80

SECTION 1 — 25 marks

Attempt ALL questions.

SECTION 2 — 55 marks

Attempt ALL questions.

Write your answers clearly in the answer booklet provided. In the answer booklet you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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X8197611

SECTION 1 — 25 marks

Attempt ALL questions

1. Two sun loungers are shown below with product information.

Luxury sun lounger

Materials

mahogany frame

textured ABS tyre

stainless steel nuts/bolts

Additional details

four adjustable positions

self-assembly (tools included)

Price – £249.99





Folding sun lounger

Materials

tubular aluminium frame

nylon netting

polypropylene footpads and corner protectors

aluminium leg brackets

Additional details

five adjustable positions

Price – £59.99

MARKS

1. (continued)

(a) Explain why the materials chosen are suitable for the sun loungers. **6**

(*You must give **six** different explanations*.)

(b) Name **three** appropriate manufacturing processes used in the production of the sun loungers **and** explain why each one is suitable. **6**

(c) Describe how function has influenced the design of the sun loungers. **5**

(d) Describe how anthropometrics and physiology have influenced the design of the sun loungers. **4**

(e) Explain the benefits of using computer aided design (CAD) during the design of the sun loungers. **4**

[Turn over

MARKS

SECTION 2 — 55 marks

Attempt ALL questions

2. A children's play frame is shown below.



The slide and side panels of the play frame are rotational moulded.

(a) Explain why rotational moulding is a suitable process for these parts. **2**

(b) State the name of a suitable material for these parts **and** explain why this material is appropriate. **3**

It is important that products are environmentally friendly.

(c) Outline steps that may have been taken to reduce the negative impact of products on the environment. **4**

MARKS

3. The success of a product relies on effective market research.

(a) Describe the key stages of **two** research techniques. **4**

Different types of specification are produced for a variety of reasons.

(b) Outline the type of information that may be contained in **two** of the following specification types:

- product design specification
- performance specification
- technical specification. **4**

4. During the design process, designers will use a variety of graphics and physical models.

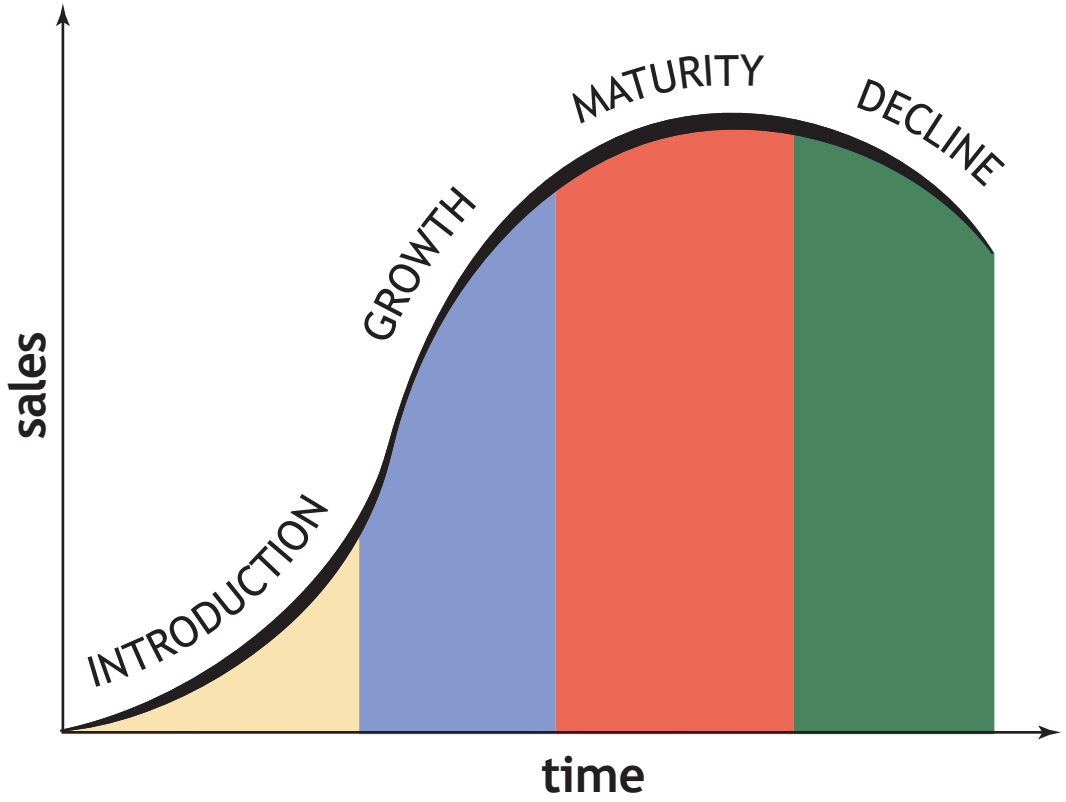
(a) Describe how graphics could be used during the design process. **4**

(b) Describe how physical models could be used during the design process. **4**

[Turn over

MARKS

5. A product life cycle graph is shown below.



During the introduction stage, marketing techniques are used.

(a) Describe marketing techniques that could be used to launch new products. **4**

A strong brand image can increase sales.

(b) Explain why a strong brand image can lead to increased sales. **3**

At the end of the product life cycle, sales will begin to decline.

(c) Explain the factors that may cause product sales to decline. **3**

Copyright is a method of protecting intellectual property.

(d) Identify another method of protecting intellectual property **and** give an example of what it protects. **2**

MARKS

6. There is often a need to identify the materials used in the manufacture of products.

(a) Describe methods that could be used to identify the materials used in a product. **4**

(*You should reference different materials in your answer.*)

Designers must ensure products can be assembled accurately and efficiently.

(b) Describe how manufacturing features could be used to ensure accurate and efficient assembly. **2**

Products can be manufactured using automated production methods.

(c) Describe how automated production methods could impact the workforce and society. **4**

7. Production and planning systems are used in the development and manufacture of new products.

Explain why production and planning systems are used. **8**

(*You should refer to a range of different production and planning systems in your answer.*)

[END OF QUESTION PAPER]

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Question 1 (Luxury sun lounger) Image of 'Berkfield Sun Lounger Solid Acacia Wood.'

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